

# RESEARCH RESULTS

## IMPACTS



### Early Warning Saves Lives

Real-time alerts allow communities to evacuate and take action before landslides strike, preventing loss of life.



### Empowers Local Communities

Provides residents direct risk updates, reducing reliance on central alerts and boosting preparedness.

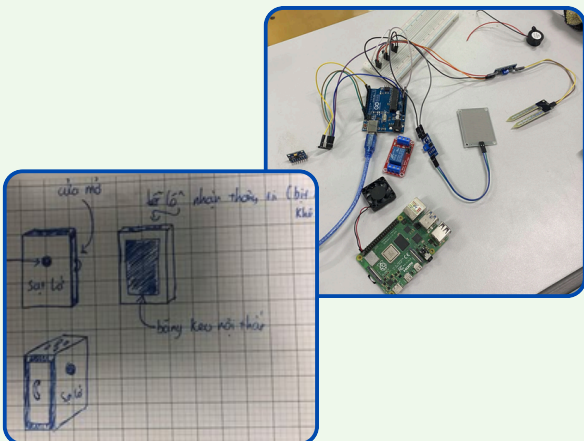


### Support for Disaster Management

Delivers crucial data, risk maps, and monitoring tools to help local authorities plan and respond more effectively.

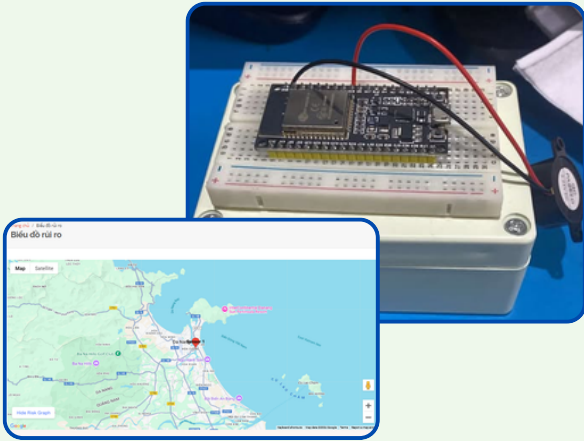
## COMPARISON OF PRODUCT VERSIONS

### Version 1



- **Components:** Sensors, Arduino Uno R3, Raspberry Pi 4
- **Function:** Detection only - no alerts, no website
- **Limitations:** Could not withstand harsh environments, no communication system, and high cost due to Raspberry Pi

### Version 2



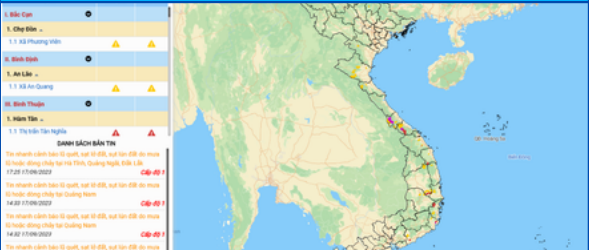
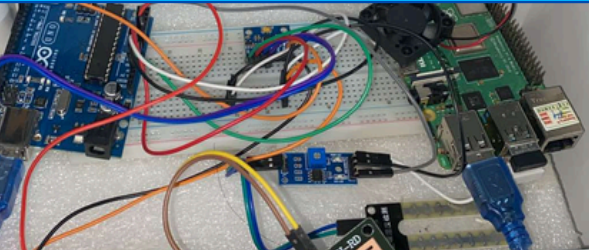
- **Improvements:** Added protective casing for sensor devices
- **Introduced:** Basic alert system and web platform (landslide map, risk charts)
- **Limitation:** Depended on Wi-Fi — limited deployment range in remote areas

### Version 3 (Current)



- **Upgraded:** Replaced Raspberry Pi with ESP32 for cost efficiency
- **Added:** Protective case for alert devices
- **Communication:** Switched to LoRa - long-range, low-power transmission
- **Web Platform:** Expanded with real-time management pages and device monitoring

## COMPARISON WITH OTHER PRODUCTS

| Names     | Student Research Topic (2011-2015)                                                  | Current Vietnam System                                                               | Our Product                                                                           |
|-----------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Products  |  |  |  |
| Sensor    | Accelerometer sensor                                                                | Based on rainfall                                                                    | Based on multiple sensors                                                             |
| Detection | Low                                                                                 | High                                                                                 | High                                                                                  |
| Warning   | Send warning to authorities                                                         | No direct warning                                                                    | Direct warning to both people and government                                          |
| Website   | None                                                                                | Shows landslide risk areas                                                           | Provides landslide risk and area, other information                                   |